STARKTECH FORENSIC REPORT

End-to-End Cyber Incident Reconstruction & Threat Analysis | APT41 Intrusion

TARGET NETWORK	THREAT ACTOR	INCIDENT DATE	PRIMARY ENTRY HOST
StarkTech Enterprise	**APT41 (Double Dragon)**	**August 25, 2025**	**DESKTOP (10.10.5.62)**

1. Executive Summary & Comprehensive Overview

On August 25, 2025, StarkTech Security Operations Center (SOC) identified multi-stage malicious activity spanning several key enterprise assets: host workstation DESKTOP (10.10.5.62), core internal server FILES-SERVER (10.10.5.245), and Domain Controller DC01. Comprehensive forensic analysis of Splunk SIEM telemetry revealed that advanced persistent threat group **APT41** gained initial access through drive-by browser activity, subsequently executing fileless PowerShell payloads, deploying redundant Remote Monitoring and Management (RMM) software, dumping domain credentials, moving laterally across subnets, and exfiltrating sensitive organizational data.

2. Complete Incident Timeline Across Attack Stages

Time (UTC)	Host & IP	Kill Chain Phase	Exploitation Details & Artifacts
13:27:40	DESKTOP (10.10.5.62)	Phase 1: Recon & Initial Access	User visited pastebin service `paste.sh` via Edge (`msedge.exe`). Sysmon EventCode 22 captured DNS resolution.
13:28:15	DESKTOP (10.10.5.62)	Phase 2: Execution & Payload Staging	PowerShell executed malicious download script, pulling `iexploreplugin.exe` from `http://10.10.5.171:8883` to disk.
13:30:56	DESKTOP (10.10.5.62)	Phase 3: Command & Control (C2)	`iexploreplugin.exe` initiated outbound connection to C2 host on port 8080 (Sysmon EventCode 3).
13:32:19	DESKTOP (10.10.5.62)	Phase 4: Discovery & Reconnaissance	Attacker issued `net user /domain` and ran SharpHound to dump AD domain telemetry to JSON files in `C:\Users\Public`.
13:35:46	DESKTOP (10.10.5.62)	Phase 5: Process Injection & Evasion	`iexploreplugin.exe` injected into `notepad.exe` (Sysmon EventCode 10) to hide execution context.
13:46:21	DESKTOP (10.10.5.62)	Phase 6: Persistence (RMM Installation)	Installed AteraAgent, Splashtop Remote, and AnyDesk services (Sysmon EventCode 7045 & Security EventCode 4697).
13:53:39	DESKTOP (10.10.5.62)	Phase 7: Defense Evasion	Executed hidden PowerShell command `Set-MpPreference -DisableRealtimeMonitoring $true` to blind Defender.
13:55:21	DESKTOP (10.10.5.62)	Phase 8: Credential Access	Opened handle to `lsass.exe` with access rights `0x1ffff` / `0x1410` to dump domain admin credentials.
14:01:29	FILES-SERVER (10.10.5.245)	Phase 9: Lateral Movement	Authenticated using stolen creds (`admin143`). Unpacked `python.zip` containing DLL side-loading payloads.
14:03:51	FILES-SERVER (10.10.5.245)	Phase 10: Payload Staging on Server	Extracted and staged `ws2_32.exe` and `system_module.exe` into Public Downloads directory.
14:17:04	FILES-SERVER (10.10.5.245)	Phase 11: Collection & Exfiltration	Compressed Finance/HR CSV files into `teams-skartech.zip`; transferred outbound via `rclone`.
14:46:02	DC01 (10.10.5.10)	Phase 12: Domain Persistence	Modified Winlogon registry keys via batch script `dhsf82.bat` setting AutoAdminLogon for account `Main\william`.

3. Phase-by-Phase Technical Incident Investigation

Phase 1: Initial Access & Weaponized Payload Delivery

The attack chain began at 13:27:40 UTC on host DESKTOP (10.10.5.62). The user navigated to `paste.sh` via Microsoft Edge. Immediately following the web request, a background PowerShell execution fetched the primary payload `iexploreplugin.exe` from an external server (`10.10.5.171:8883`), masquerading as an Internet Explorer plugin to avoid scrutiny.

Figure 1 & 2: Web Reconnaissance & Initial Query to paste.sh — `Sysmon EventCode 22 (DNS)`

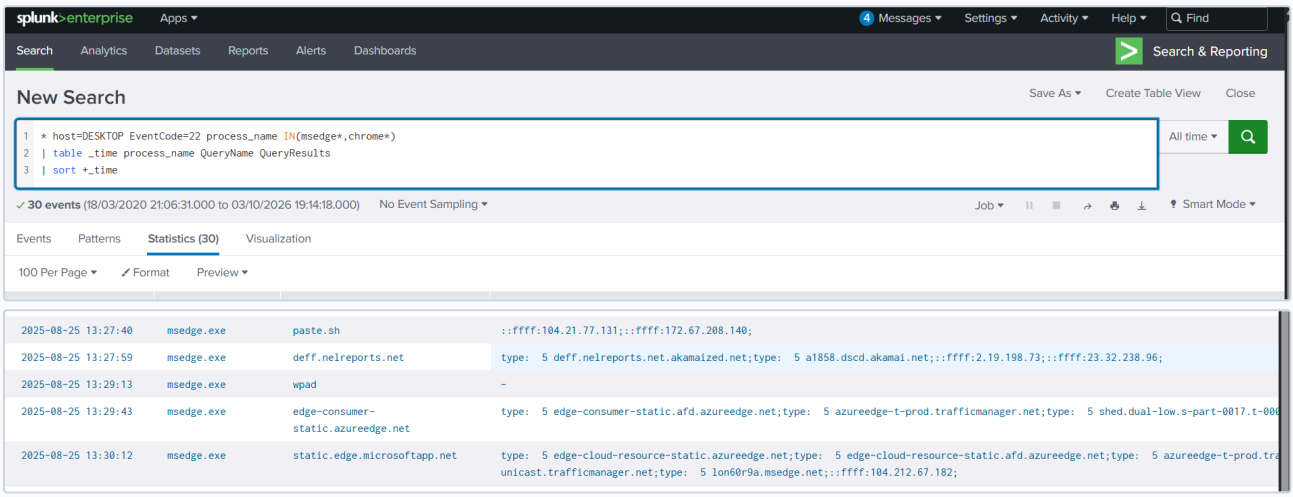
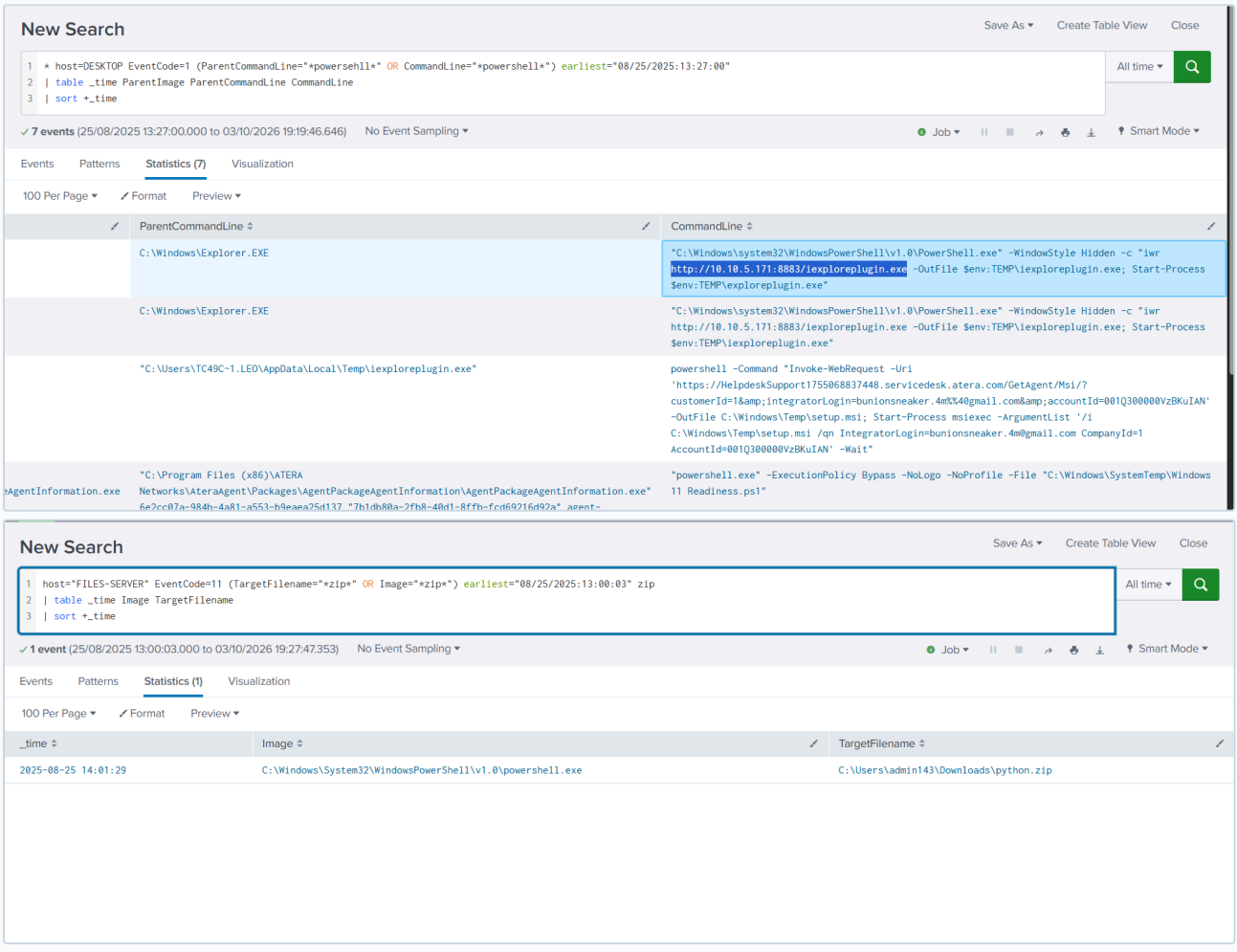


Figure 3 & 4: PowerShell Stage-1 Downloader & File Creation — `Sysmon EventCode 1 / EventCode 11`



Phase 2: Lateral Movement Prep & Python DLL Side-Loading Staging

Upon establishing foothold, APT41 expanded execution to `FILES-SERVER` (10.10.5.245). The adversary transferred a compressed package `python.zip`, extracting Python binaries along with custom DLL side-loading runners (`testc.exe`, `python311.dll`). This environment allowed them to execute arbitrary Python scripts while bypassing standard application whitelisting policies.

Figure 5 & 6: Archive Extraction via Expand-Archive on Server `PowerShell Expand-Archive`

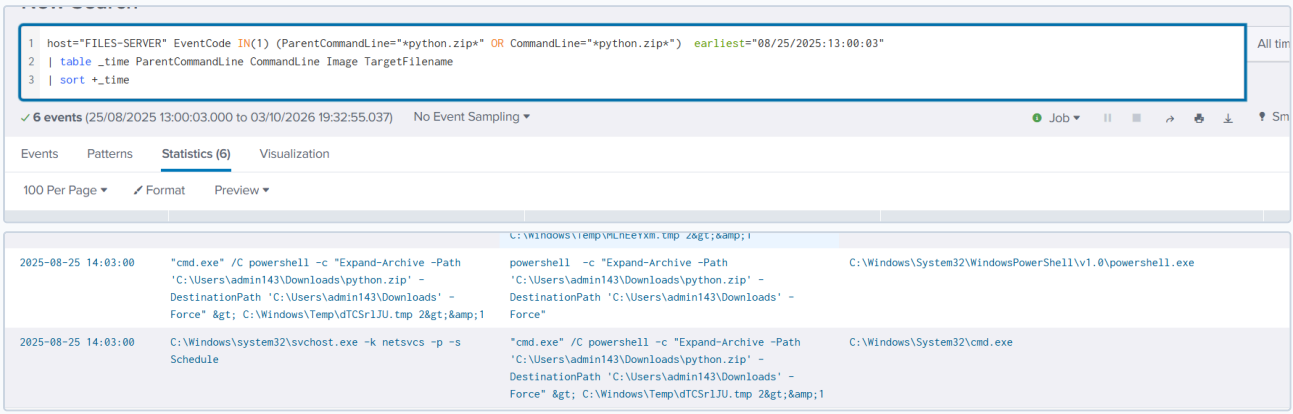
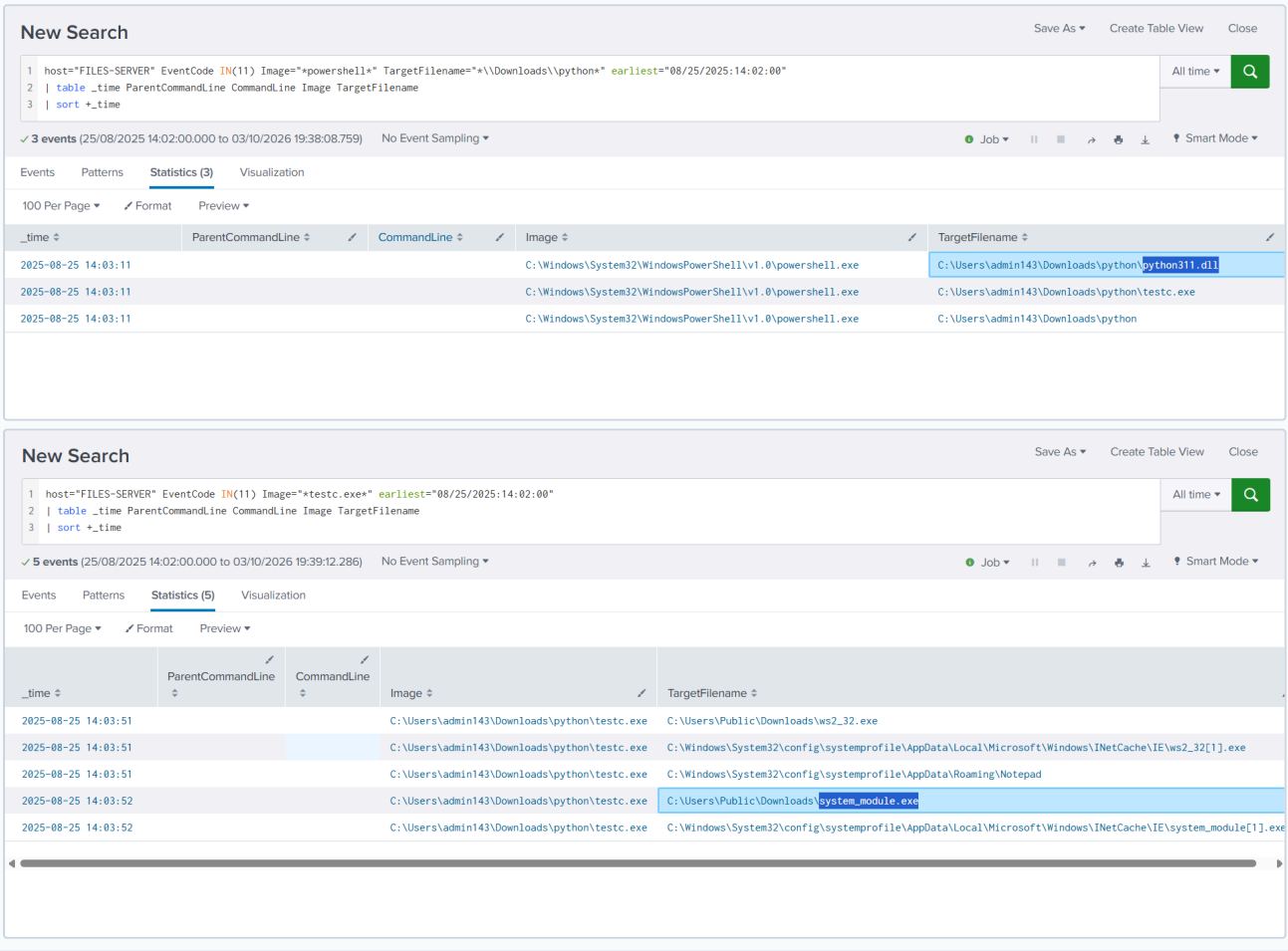


Figure 7 & 8: Extraction of testc.exe & Staging Executables in Downloads `Sysmon EventCode 11 (FileCreate)`



Phase 3: Command & Control Channel Establishment & Traffic Analysis

Following local execution, `iexploreplugin.exe` established a continuous beaconing session back to external IP infrastructure on port 8080. Network flow logs and Suricata NIDS alerts confirmed HTTP traffic containing encrypted payload heartbeats and tasking instructions coming directly from `DESKTOP` and `FILES-SERVER`.

Figure 9 & 10: Beaconing C2 Network Traffic & HTTP Host Logs `Sysmon EventCode 3 / Suricata eve.json`

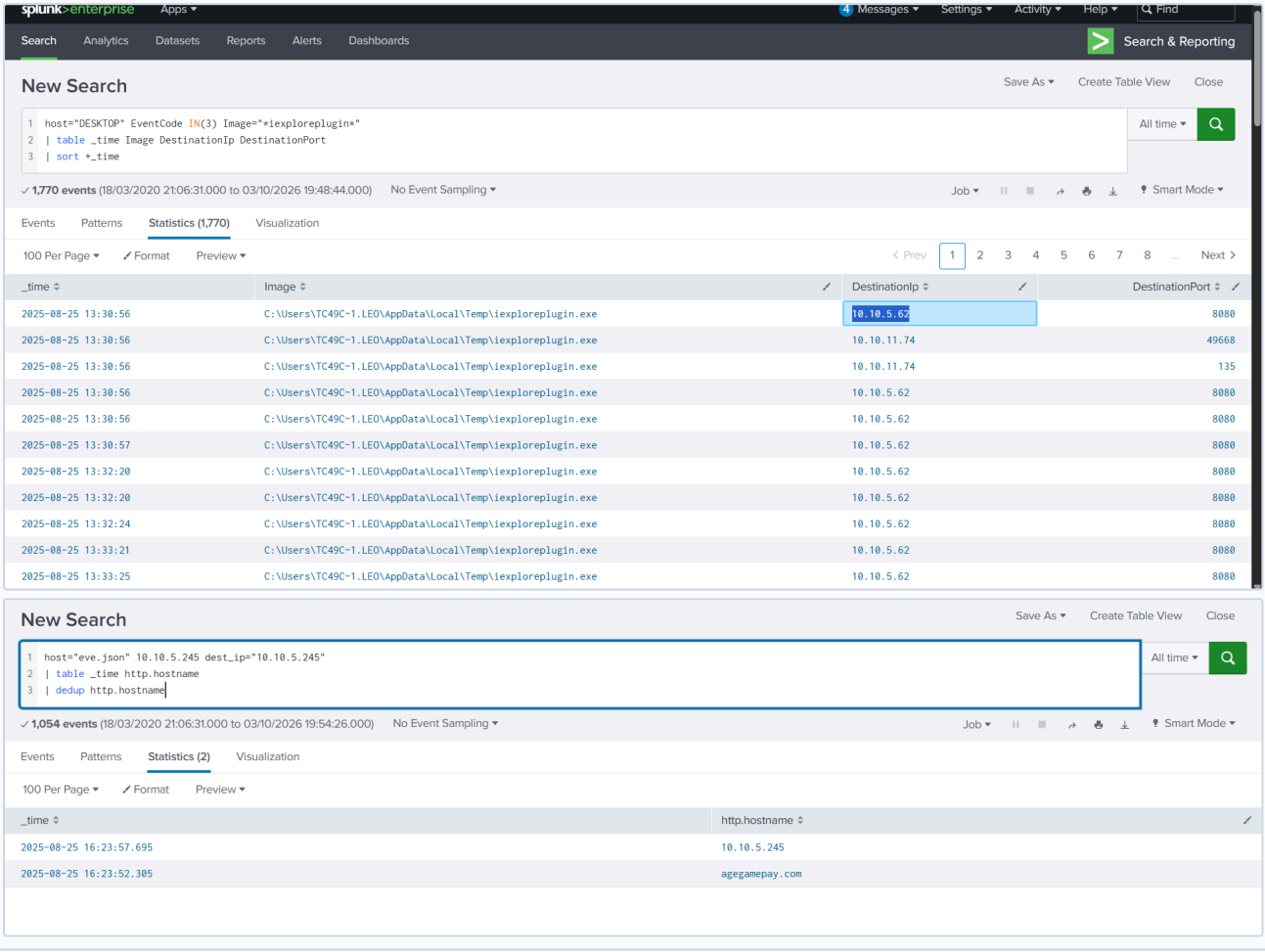
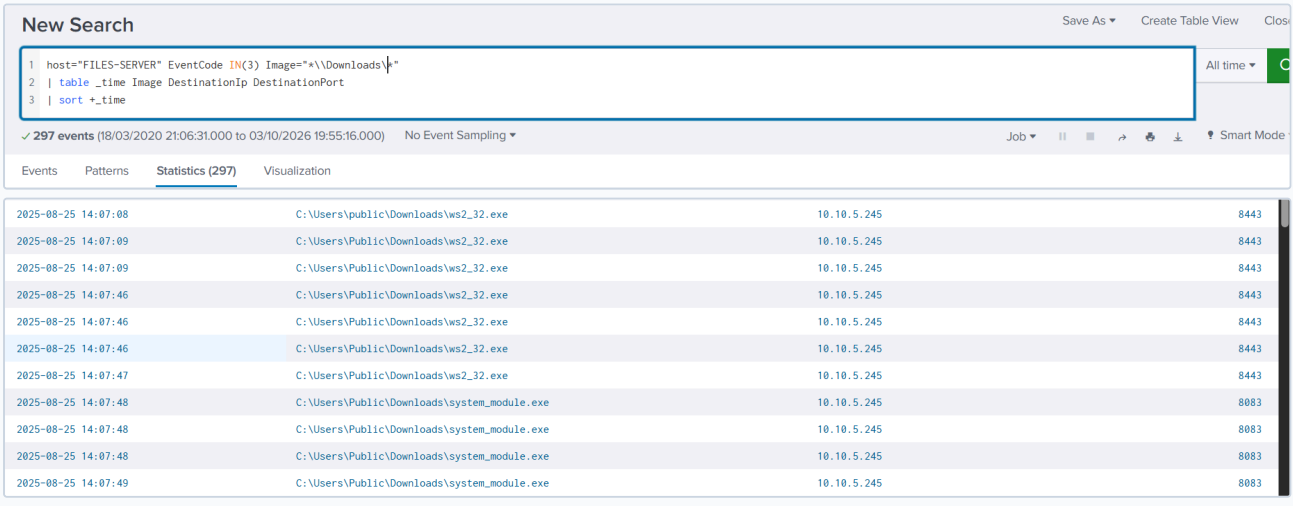


Figure 11 & 12: Secondary Outbound Channels & Execution from Public Directories `DestinationPorts 8083 / 8443`



Phase 4: Persistence via Remote Monitoring Tools & Scheduled Tasks

To guarantee permanent access independent of C2 implants, APT41 installed legitimate Remote Monitoring and Management (RMM) software. Using `msiexec`, they deployed Atera Agent, Splashtop, and AnyDesk as background services. They also created scheduled tasks linked to `svchost.exe` to maintain secondary triggers.

Figure 13 & 14: MSI Installer Execution & Atera Remote Software Installation `msiexec.exe / setup.msi`

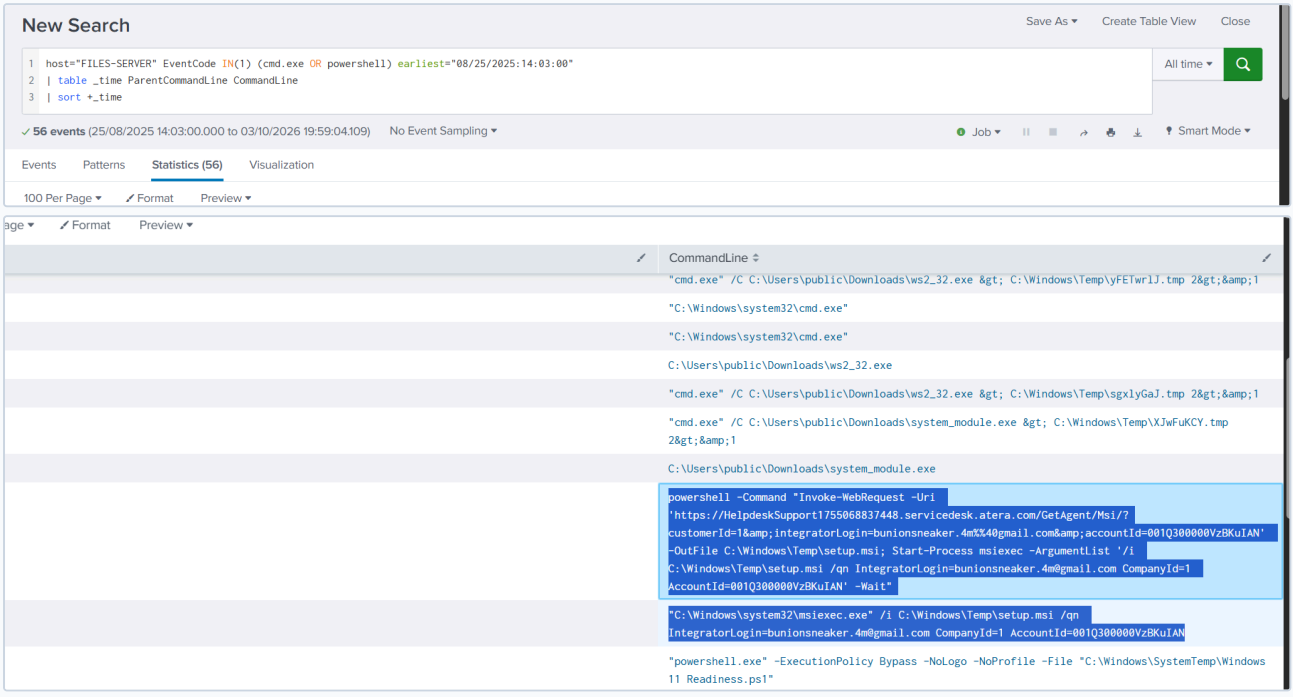


Figure 15 & 16: System Service Creation Logs (Atera, Splashtop, AnyDesk) `Sysmon EventCode 7045 / Security EventCode 4697`

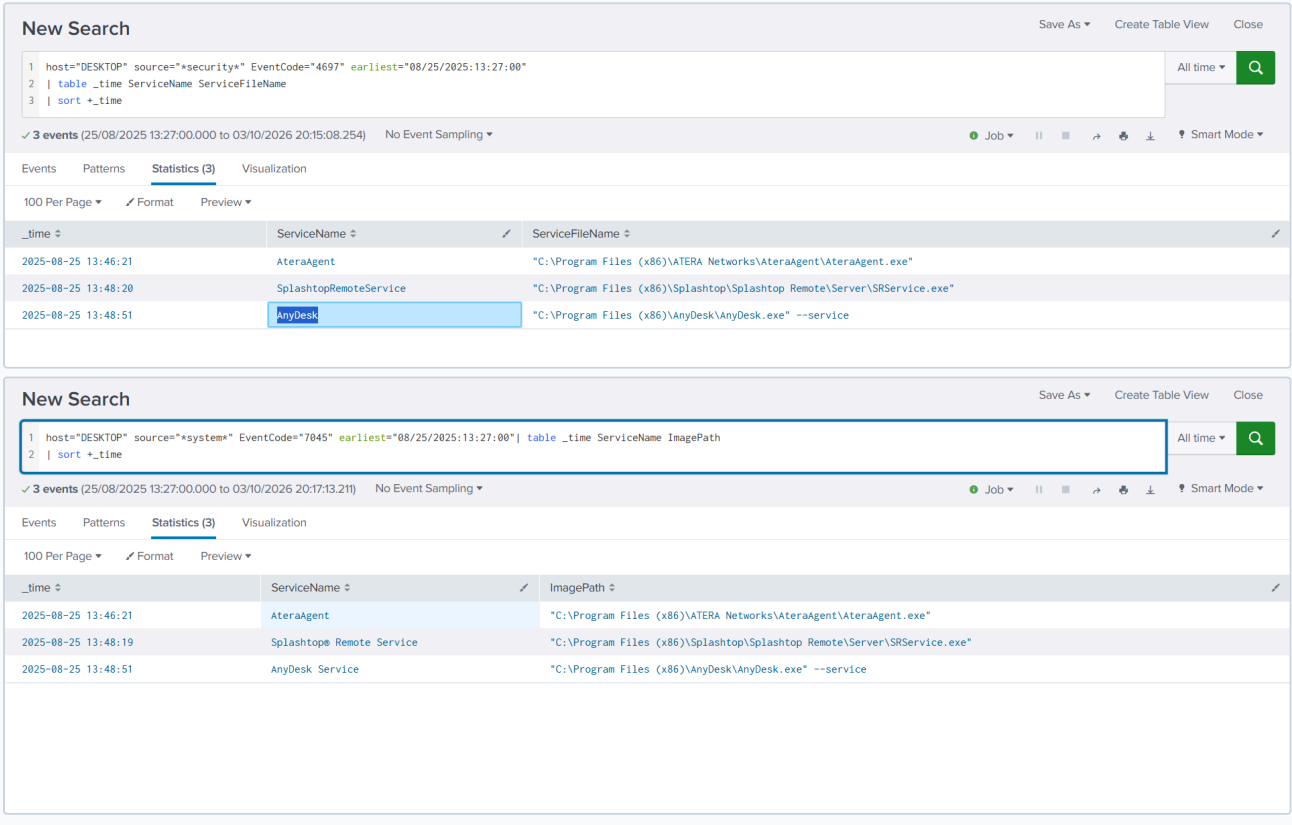
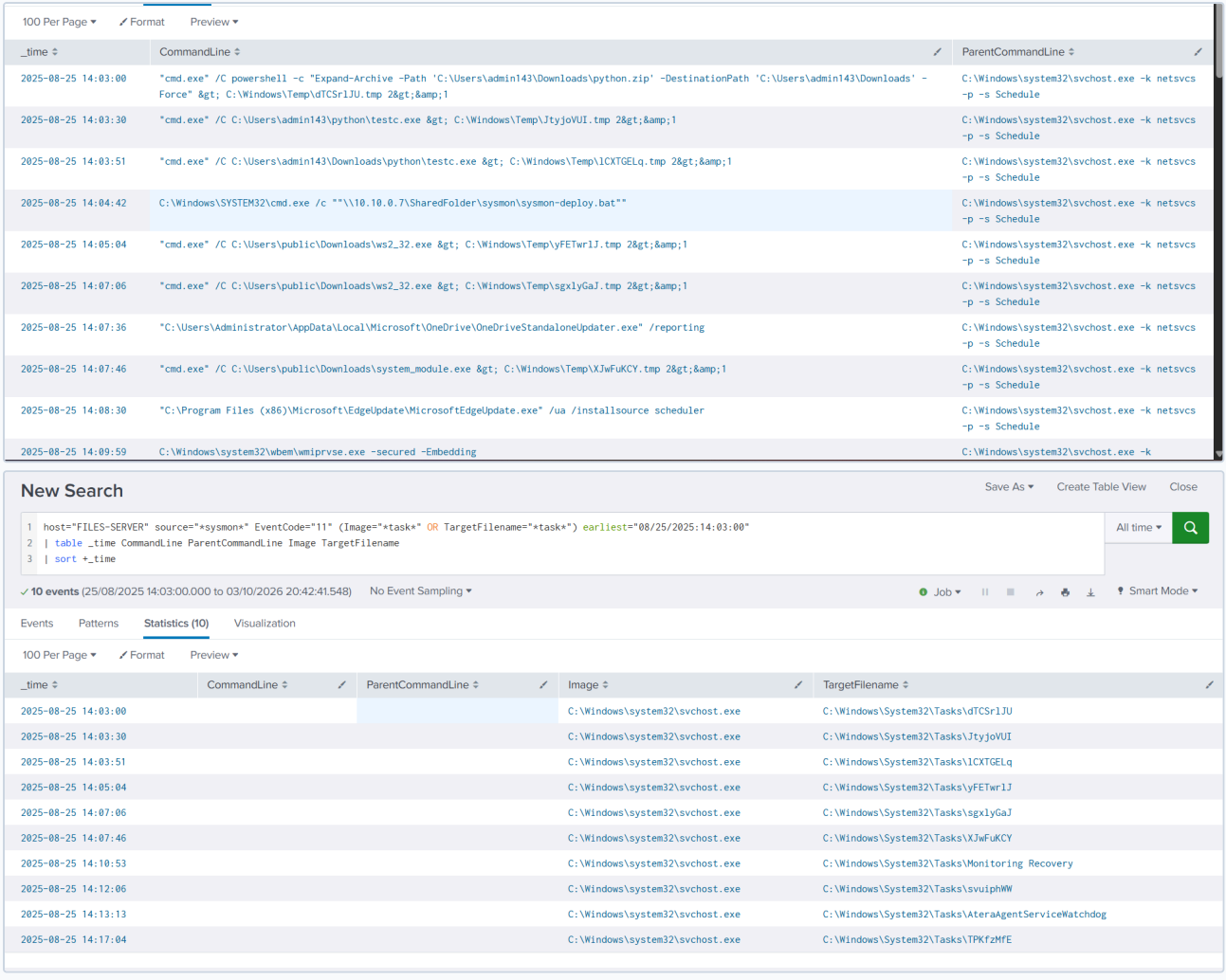


Figure 17 & 18: Scheduled Task Registration via Task Scheduler Service `svchost.exe -k netsvcs -p -s Schedule`



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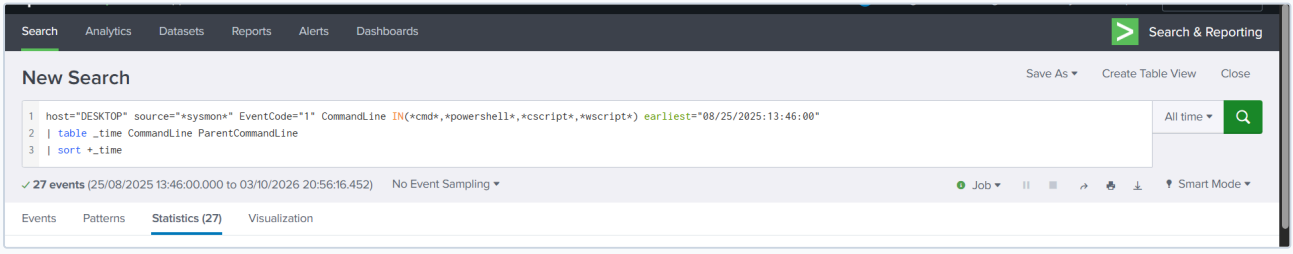
_time	CommandLine	ParentCommandLine
2025-08-25 14:03:00	"cmd.exe" /C powershell -c "Expand-Archive -Path 'C:\Users\admin143\Downloads\python.zip' -DestinationPath 'C:\Users\admin143\Downloads' -Force" > C:\Windows\Temp\dTCSr1JU.tmp 2>&1	C:\Windows\system32\svchost.exe -k netsvcs -p -s Schedule
2025-08-25 14:03:30	"cmd.exe" /C C:\Users\admin143\python\testc.exe > C:\Windows\Temp\JtyjoVUI.tmp 2>&1	C:\Windows\system32\svchost.exe -k netsvcs -p -s Schedule
2025-08-25 14:03:51	"cmd.exe" /C C:\Users\admin143\Downloads\python\testc.exe > C:\Windows\Temp\lCXTGELq.tmp 2>&1	C:\Windows\system32\svchost.exe -k netsvcs -p -s Schedule
2025-08-25 14:04:42	C:\Windows\SYSTEM32\cmd.exe /c ""\\10.10.0.7\SharedFolder\sysmon\sysmon-deploy.bat""	C:\Windows\system32\svchost.exe -k netsvcs -p -s Schedule
2025-08-25 14:05:04	"cmd.exe" /C C:\Users\public\Downloads\ws2_32.exe > C:\Windows\Temp\yFETwr1J.tmp 2>&1	C:\Windows\system32\svchost.exe -k netsvcs -p -s Schedule
2025-08-25 14:07:06	"cmd.exe" /C C:\Users\public\Downloads\ws2_32.exe > C:\Windows\Temp\sgxlyGaJ.tmp 2>&1	C:\Windows\system32\svchost.exe -k netsvcs -p -s Schedule
2025-08-25 14:07:36	"C:\Users\Administrator\AppData\Local\Microsoft\OneDrive\OneDriveStandaloneUpdater.exe" /reporting	C:\Windows\system32\svchost.exe -k netsvcs -p -s Schedule
2025-08-25 14:07:46	"cmd.exe" /C C:\Users\public\Downloads\system_module.exe > C:\Windows\Temp\XJwFuKCY.tmp 2>&1	C:\Windows\system32\svchost.exe -k netsvcs -p -s Schedule
2025-08-25 14:08:30	"C:\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe" /ua /installsource scheduler	C:\Windows\system32\svchost.exe -k netsvcs -p -s Schedule
2025-08-25 14:09:59	C:\Windows\system32\wbem\wmiprvse.exe -secured -Embedding	C:\Windows\system32\svchost.exe -k

_time	CommandLine	ParentCommandLine	Image	TargetFilename
2025-08-25 14:03:00			C:\Windows\system32\svchost.exe	C:\Windows\System32\Tasks\dTCSr1JU
2025-08-25 14:03:30			C:\Windows\system32\svchost.exe	C:\Windows\System32\Tasks\JtyjoVUI
2025-08-25 14:03:51			C:\Windows\system32\svchost.exe	C:\Windows\System32\Tasks\lCXTGELq
2025-08-25 14:05:04			C:\Windows\system32\svchost.exe	C:\Windows\System32\Tasks\yFETwr1J
2025-08-25 14:07:06			C:\Windows\system32\svchost.exe	C:\Windows\System32\Tasks\sgxlyGaJ
2025-08-25 14:07:46			C:\Windows\system32\svchost.exe	C:\Windows\System32\Tasks\XJwFuKCY
2025-08-25 14:10:53			C:\Windows\system32\svchost.exe	C:\Windows\System32\Tasks\Monitoring Recovery
2025-08-25 14:12:06			C:\Windows\system32\svchost.exe	C:\Windows\System32\Tasks\svuiphWW
2025-08-25 14:13:13			C:\Windows\system32\svchost.exe	C:\Windows\System32\Tasks\AteraAgentServiceWatchdog
2025-08-25 14:17:04			C:\Windows\system32\svchost.exe	C:\Windows\System32\Tasks\TPKfzMfE

Phase 5: Defense Evasion, Registry Tampering & Domain Reconnaissance

To evade endpoint security, the attacker executed VBS scripts (`scvhost.vbs`) and ran PowerShell commands disabling Windows Defender Realtime Monitoring. On Domain Controller DC01, they modified Winlogon registry keys (`AutoAdminLogon=1`) to ensure automatic domain admin access upon reboot.

Figure 19 & 20: Disabling Defender Realtime Protection via Scripting `Set-MpPreference -DisableRealtimeMonitoring`



_time	CommandLine	ParentCommandLine
	"C:\Windows\system32\wevtutil.exe" um "C:\ProgramData\Splashtop\Common\Event\stevt_srs_provider.man"	
2025-08-25 13:48:16	"C:\Windows\sysnative\cmd.exe" /C "C:\Windows\system32\wevtutil.exe" im "C:\ProgramData\Splashtop\Common\Event\stevt_srs_provider.man"	"C:\Program Files (x86)\Splashtop\Splashtop Remote\Server\Support\SetupUtil.exe" /P ST_EVENT
2025-08-25 13:48:22	cmd.exe /c "C:/ProgramData/AnyDesk.exe /S"	"C:\Users\TC49C~1.LEO\AppData\Local\Temp\iexploreplugin.exe"
2025-08-25 13:48:48	"cmd.exe" /K "cd /d C:\Program Files (x86)\ATERA Networks\AteraAgent\Packages\AgentPackageRuntimeInstaller\" /	-
2025-08-25 13:49:27	"cmd.exe" /K "cd /d C:\Program Files (x86)\ATERA Networks\AteraAgent\Packages\AgentPackageRuntimeInstaller\" /	"C:\Program Files (x86)\ATERA Networks\AteraAgent\Packages\AgentPackageRuntimeInstaller\AgentPackageRuntimeInstaller.exe" 6e "eyJDb21tYW5kTmFtZSI6Imluc3RhbGxkb3RuZXQiLCJEb3ROZXRWZXJzaW9uIjoiOC4wLjExIiwiTWFjQVJNRG93bmxvYWRVcmwiOiJodHRwczovL2Rvd25sb2F001Q300000VzBKuIAN
2025-08-25 13:49:28	"cmd.exe" /K "cd /d C:\Program Files (x86)\ATERA Networks\AteraAgent\Packages\AgentPackageRuntimeInstaller\" /	"C:\Program Files (x86)\ATERA Networks\AteraAgent\Packages\AgentPackageRuntimeInstaller\AgentPackageRuntimeInstaller.exe" 6e "eyJDb21tYW5kTmFtZSI6Imluc3RhbGxkb3RuZXQiLCJEb3ROZXRWZXJzaW9uIjoiOC4wLjExIiwiTWFjQVJNRG93bmxvYWRVcmwiOiJodHRwczovL2Rvd25sb2F001Q300000VzBKuIAN
2025-08-25 13:52:25	cscript.exe C:\Users\t.leon\scvhost.vbs	cmd.exe /c "cscript.exe C:\Users\t.leon\scvhost.vbs"
2025-08-25 13:52:25	cmd.exe /c "cscript.exe C:\Users\t.leon\scvhost.vbs"	"C:\Users\TC49C~1.LEO\AppData\Local\Temp\iexploreplugin.exe"
2025-08-25 13:53:39	cmd.exe /c "powershell.exe -WindowStyle Hidden -Command Set-MpPreference -DisableRealtimeMonitoring"	"C:\Users\TC49C~1.LEO\AppData\Local\Temp\iexploreplugin.exe"
2025-08-25 13:53:40	powershell.exe -WindowStyle Hidden -Command Set-MpPreference -DisableRealtimeMonitoring	cmd.exe /c "powershell.exe -WindowStyle Hidden -Command Set-MpPreference -DisableRealtimeMonitoring"
2025-08-25 14:04:38	C:\Windows\SYSTEM32\cmd.exe /c ""\\10.10.0.7\SharedFolder\sysmon\sysmon-deploy.bat""	C:\Windows\system32\svchost.exe -k netsvcs -p -s Schedule
2025-08-25 15:04:38	C:\Windows\SYSTEM32\cmd.exe /c ""\\10.10.0.7\SharedFolder\sysmon\sysmon-deploy.bat""	C:\Windows\system32\svchost.exe -k netsvcs -p -s Schedule

Figure 21 & 22: Winlogon Persistence Scripting & Domain Account Discovery `dhsf82.bat / net user /domain`

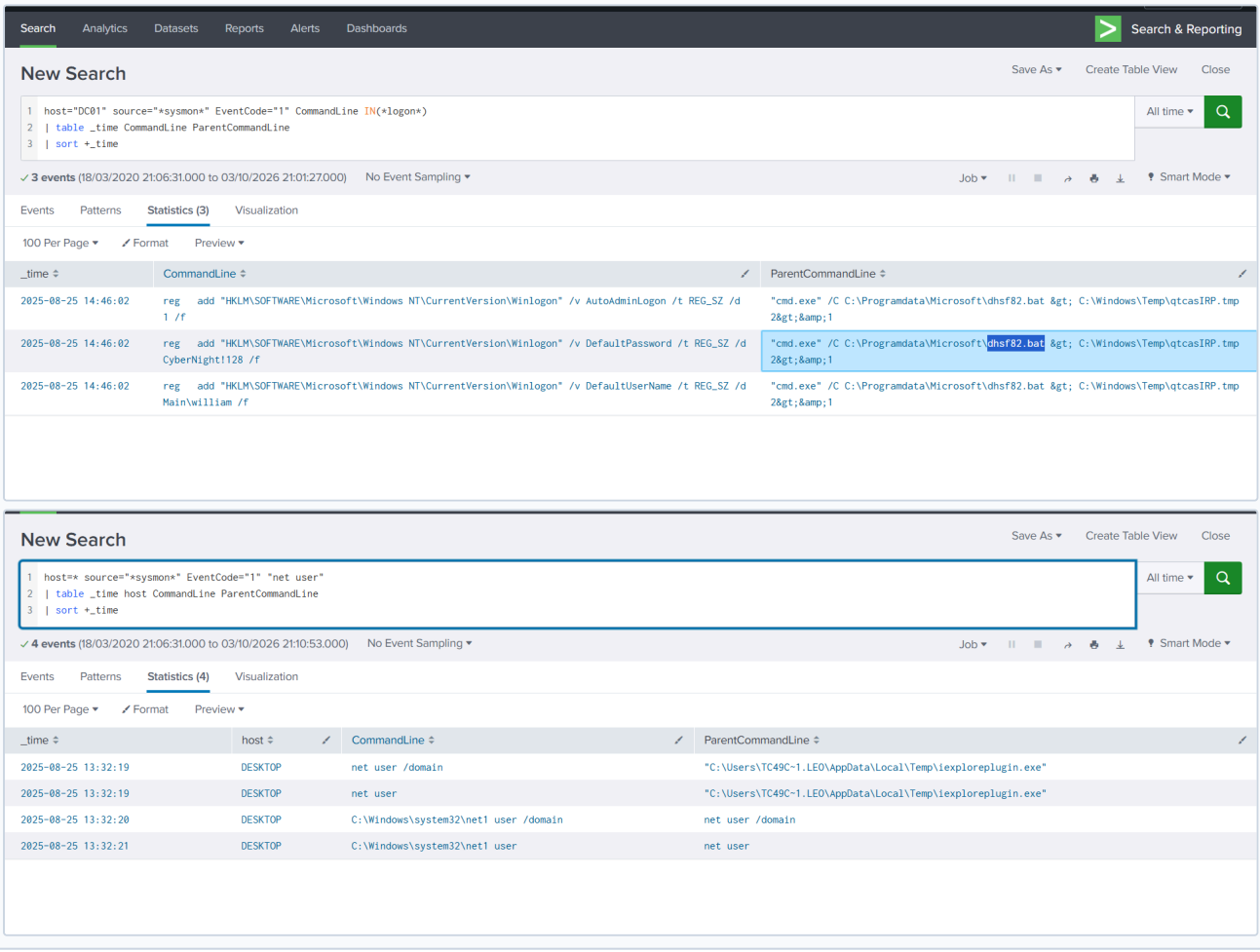
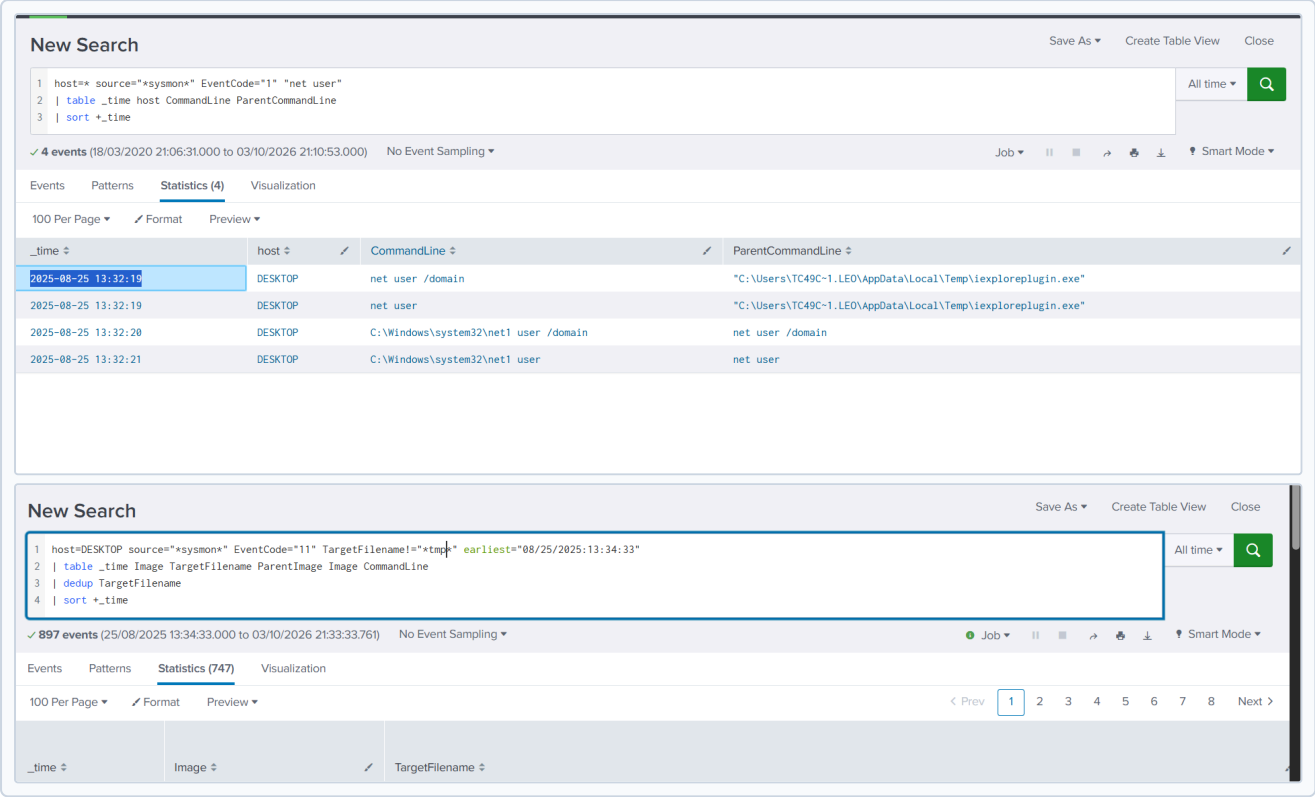


Figure 23 & 24: Active Directory Object Enumeration & File Creation Events `Sysmon EventCode 11 (File Creation)`



Phase 6: BloodHound Recon, Credential Access (LSASS) & Process Injection

Using SharpHound, APT41 collected Active Directory trust relationships, exporting JSON structure files into `C:\Users\Public`. Simultaneously, `iexploreplugin.exe` injected into `notepad.exe` and requested full process access memory handles (`0x1ffff`) to `lsass.exe` to extract plaintext passwords and NTLM hashes.

Figure 25 & 26: BloodHound Graph Data Dumping to Public Directory `BloodHound JSON Exports / Public Dir`

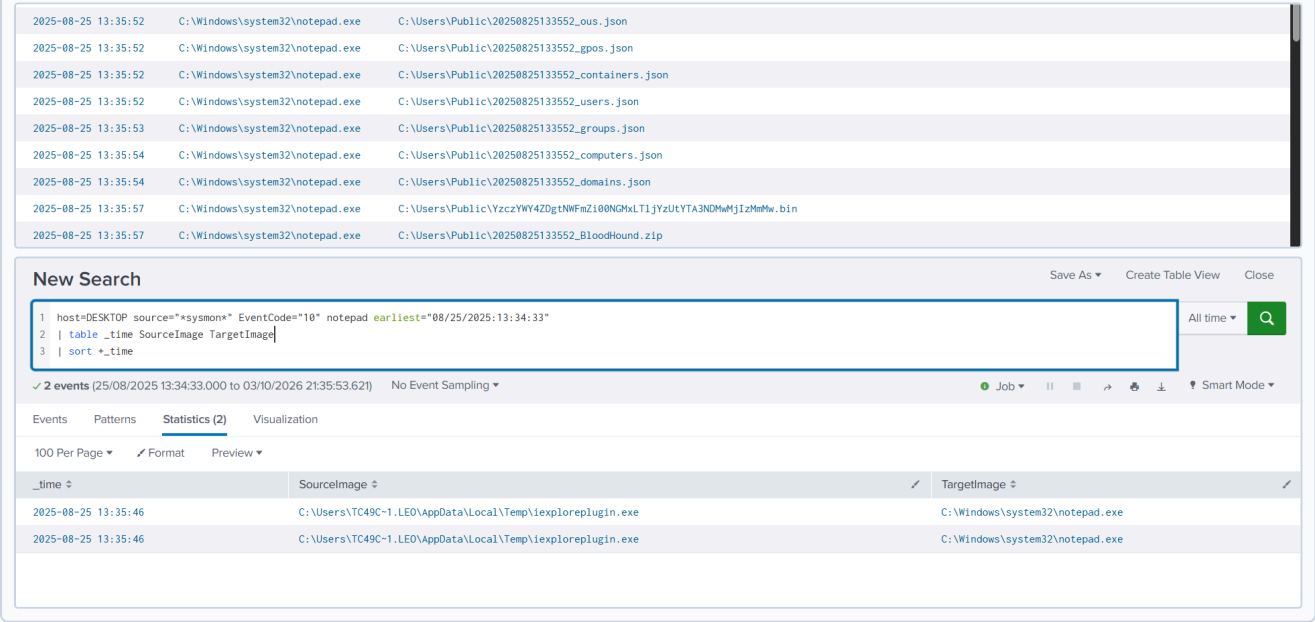


Figure 27 & 28: Notepad Process Injection & PowerShell Defender Override `Sysmon EventCode 10 (ProcessAccess)`

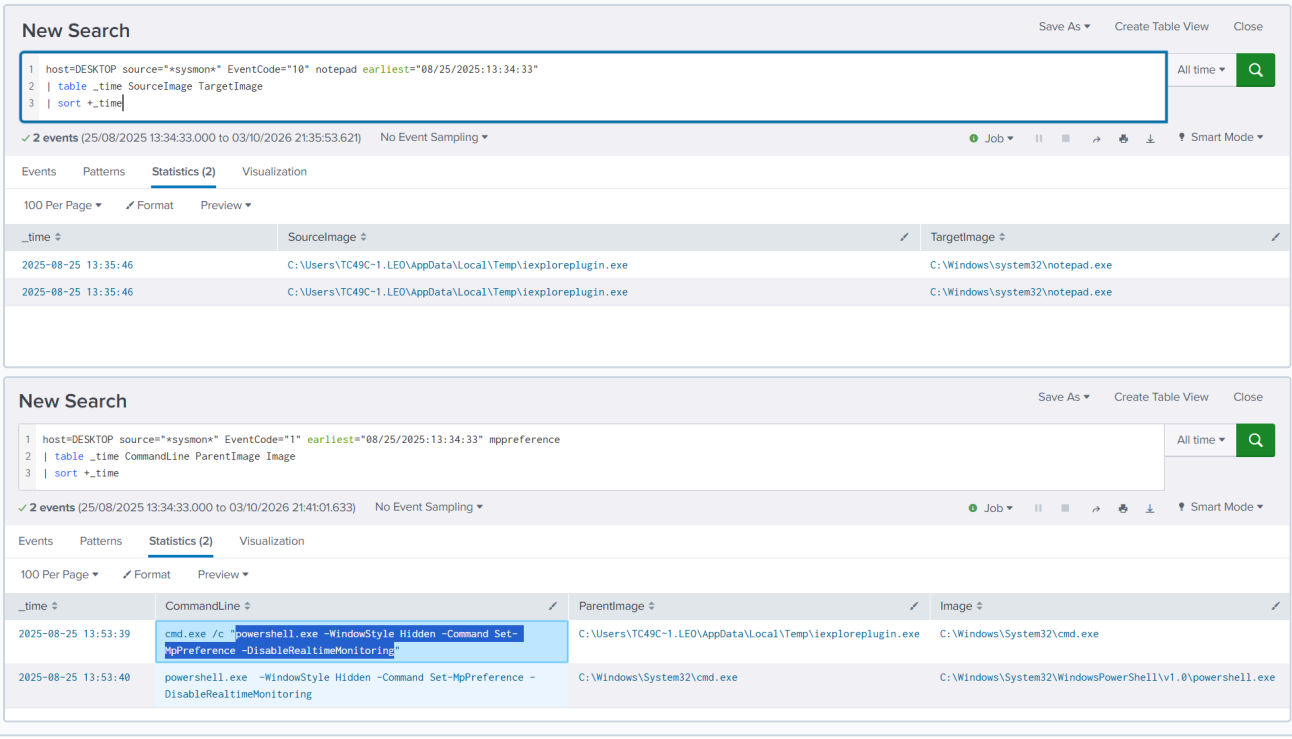
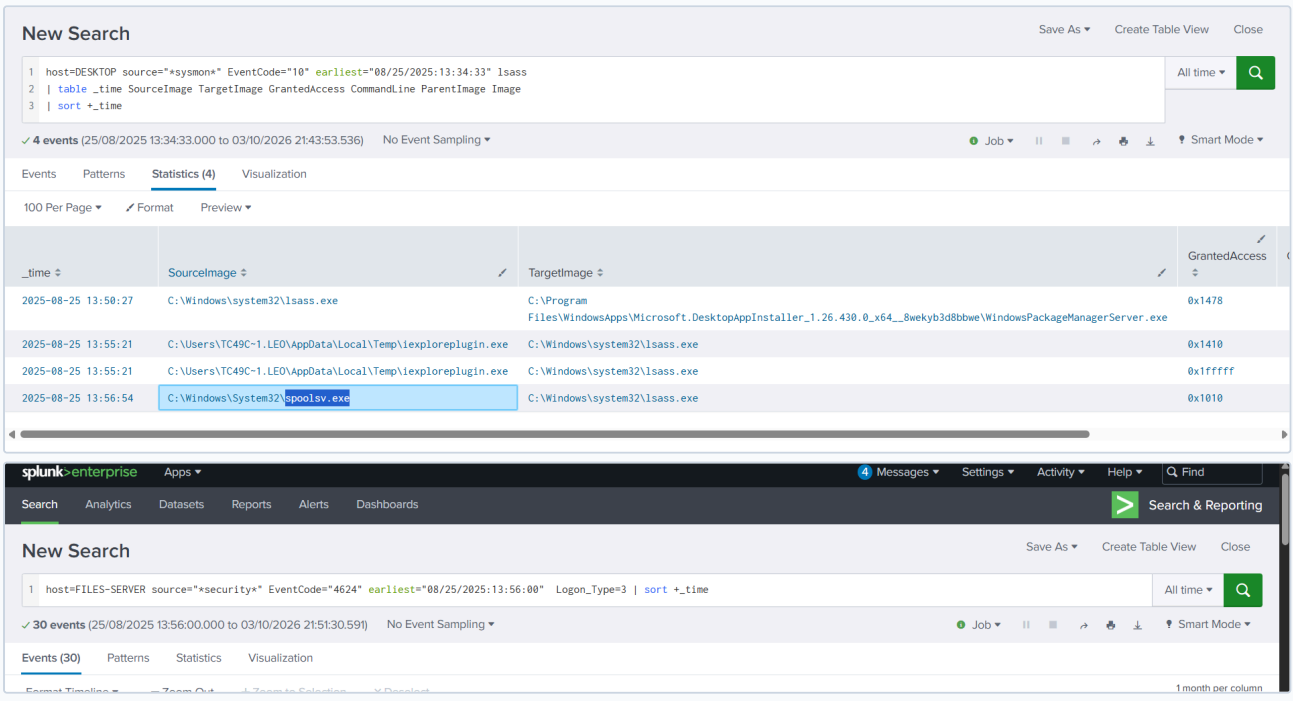


Figure 29 & 30: Direct LSASS Memory Scraping & Network Authentication `GrantedAccess 0x1ffff / Security Event 4624`



Phase 7: Account Impersonation, Data Staging & Cloud Exfiltration

Leveraging compromised domain account `CORETECH\admin143`, APT41 moved to `FILES-SERVER`, gathering high-value HR and Financial CSV files. The data was compressed into `teams-skartech.zip` and exfiltrated to an external cloud endpoint via `rclone`. Remote PowerShell scripts were fetched via `certutil` to finalize cleanup.

Figure 31 & 32: Network Logon Telemetry & Privileged User Impersonation — `LogonType 3 / CORETECH\admin143`

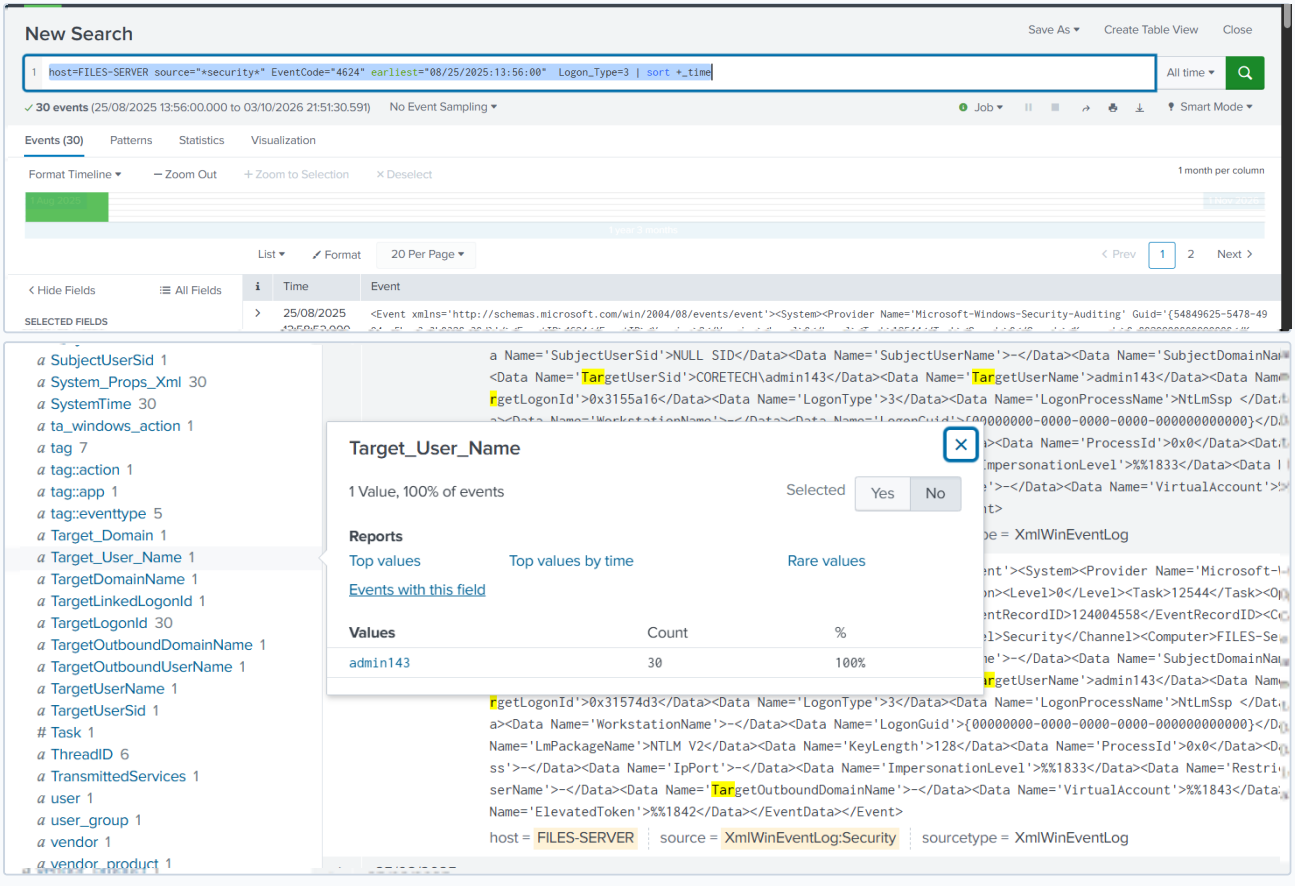
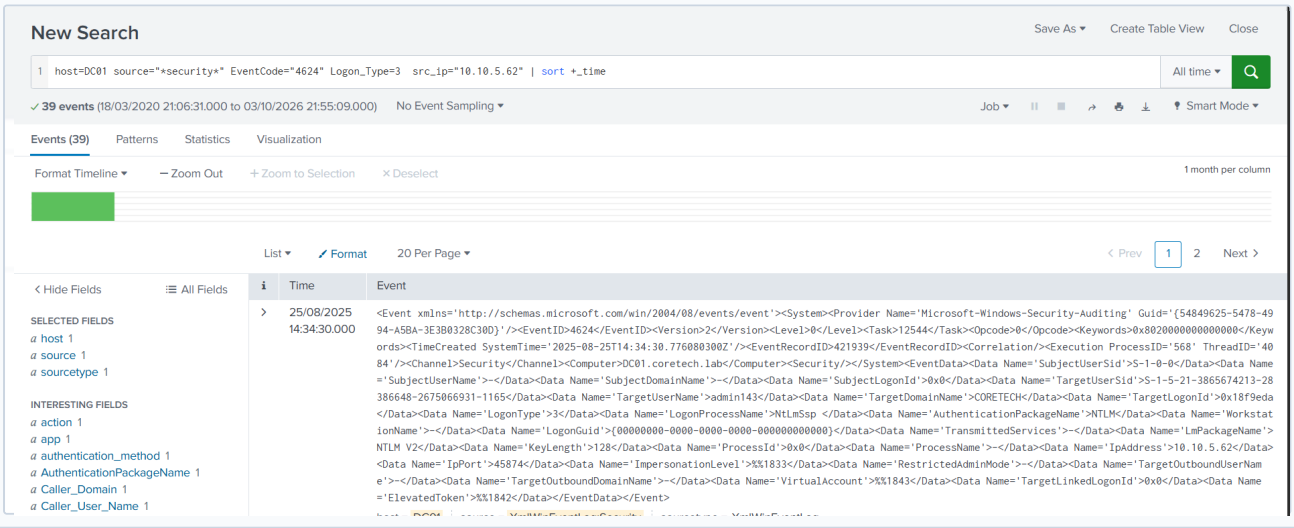


Figure 33 & 34: Domain Controller NTLM Session & Finance/HR Data Archive — `teams-skartech.zip Compression`



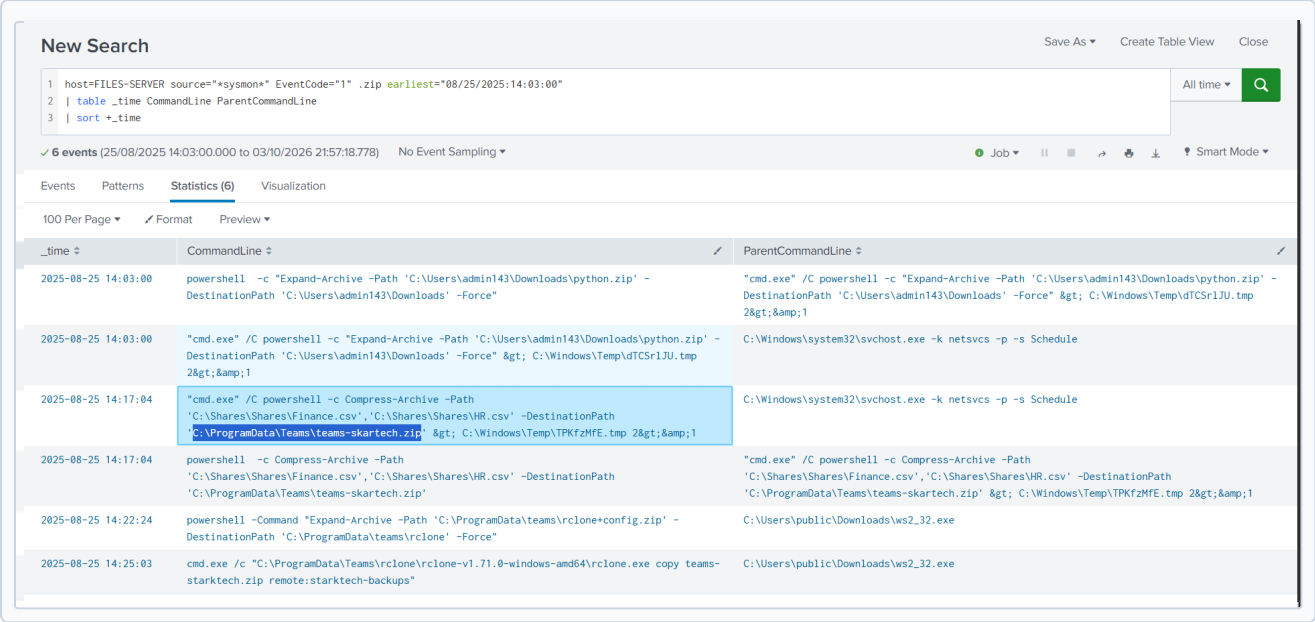
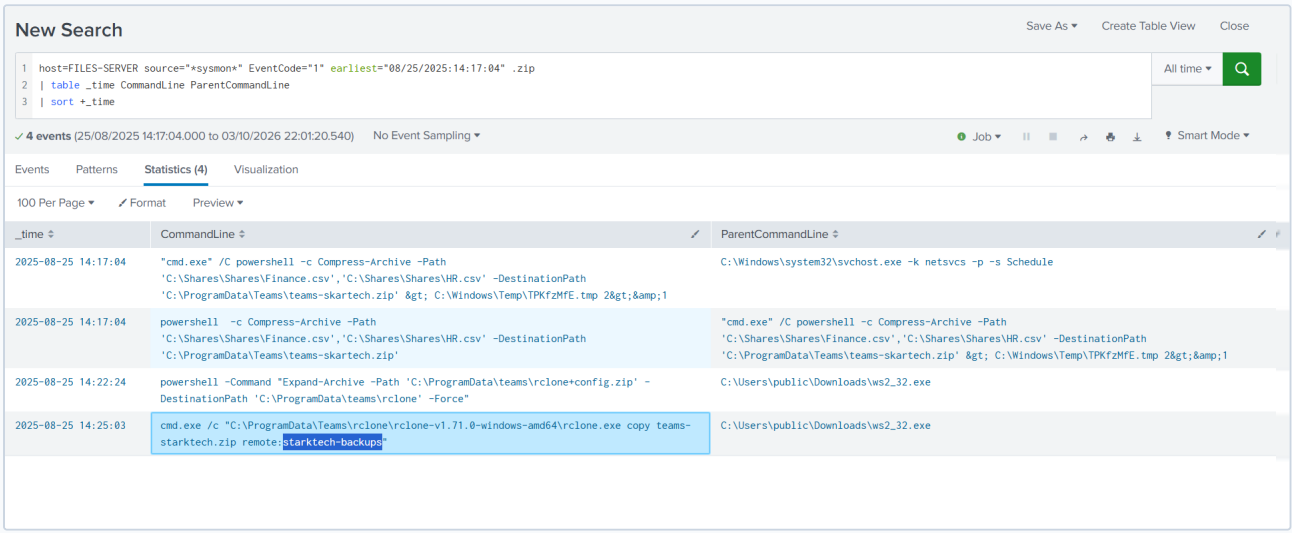
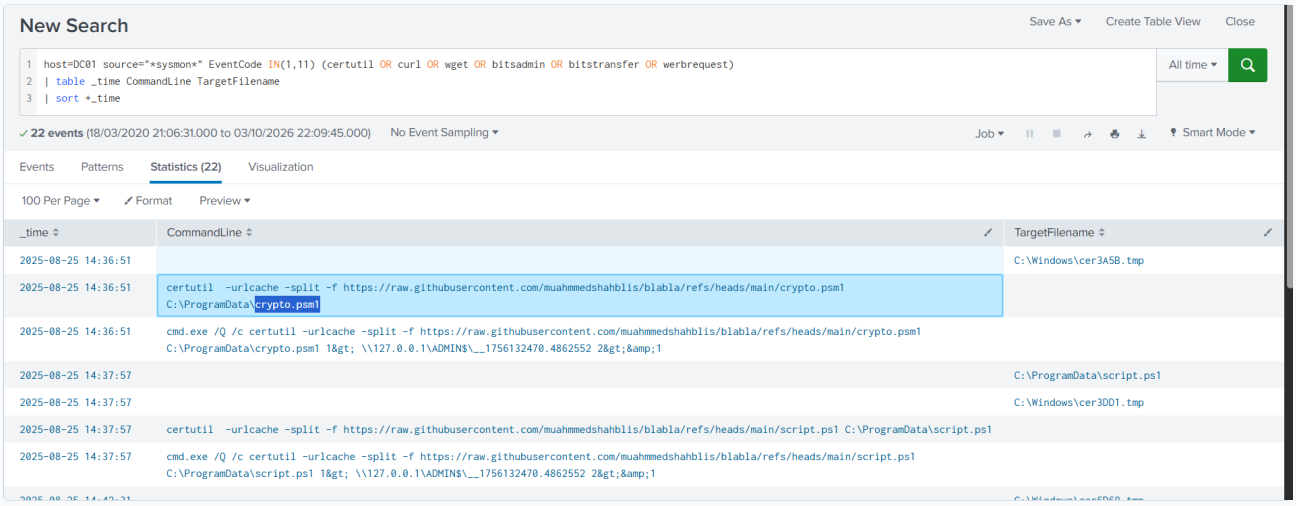


Figure 35 & 36: Cloud Exfiltration via RClone & Certutil Script Download `rclone copy / certutil -urlcache`





4. Strategic Mitigation & Remediation Roadmap

To fully neutralize threat actor persistence and mitigate future risks associated with APT41 tradecraft, the StarkTech Security Operations Center (SOC) must execute the following structured remediation plan across Immediate, Short-Term, and Long-Term phases:

Phase 1: Immediate Containment & Eradication (0-24 Hours)

Domain Area	Required Security Action	Target Host / Account
Identity Revocation	Force Global Password Reset & invalidate active Kerberos TGT tickets (double reset krbtgt).	`CORETECH\admin143`, `Main\william`
Service Termination	Stop, disable, and purge rogue RMM services: AteraAgent, SplashtopRemoteService, AnyDesk.	`DESKTOP`, `FILES-SERVER`
Registry Cleanup	Delete malicious Winlogon registry keys (`AutoAdminLogon`, `DefaultPassword`).	`DC01` (`HKLM\SOFTWARE\...`)
Task Eviction	Remove rogue Scheduled Tasks (`dTCSr1JU`, `JtyjoVUI`, `LCXTGELq`, `yFETwrlJ`, `sgxlyGaJ`).	`FILES-SERVER`

Phase 2: System Hardening & Security Controls (1-7 Days)

Control Category	Technical Control Description
LSASS Protection	Enable **LSASS RunAsPPA** (Protected Process Light) via Group Policy to block process memory access and credential harvesting attempt (`GrantedAccess 0x1ffff`).
RMM Software Control	Implement AppLocker / Windows Defender Application Control (WDAC) block rules for unauthorized remote administration software (Atera, AnyDesk, Splashtop, RClone).
Network Egress Filtering	Restrict outbound connectivity from core servers. Block non-standard egress ports (8080, 8083, 8443) and restrict DNS lookup to approved internal recursive resolvers.
PowerShell Hardening	Enforce Constrained Language Mode (CLM) and enable Script Block Logging (EventID 4104) across all domain-joined endpoints.

Phase 3: Long-Term Architecture & Monitoring Enhancements

1. **EDR Deployment & Anti-Tamper Enforcement:** Mandate Endpoint Detection and Response (EDR) policy locks preventing command-line disabling of real-time monitoring (`Set-MpPreference`).

2. **Privileged Access Management (PAM):** Tier domain administration accounts; strictly eliminate domain admin logins on standard workstation endpoints (`DESKTOP`).

3. **SIEM Rule Correlation:** Implement automated alerts combining EventCode 10 (LSASS access), EventCode 7045 (Service Installation), and Certutil HTTP downloads.